

Computational Complexity, DP, and recursion

Quiz 2 Practice

10-607

Disclaimer: This isn't exactly a "specimen paper." That is, some problems here are going to be more difficult, and some are going to be easier. However, this is a good place to start to find out which concepts you need to work on more. This also isn't meant to reflect the length or format of the actual quiz.

1. Let $A = a_1, a_2, \dots, a_N$, $B = b_1, b_2, \dots, b_N$, and $C = c_1, c_2, \dots, c_N$ be three sequences of length N . The longest common subsequence is the longest sequence $S = s_1, \dots, s_K$ such that S is a subsequence of A , B , and C . Note that a subsequence is defined as any sequence that can be formed by deleting one or more characters from a sequence. For example, "abd" and "be" are valid subsequences of the sequence "abcde". Provide a dynamic programming algorithm that, given sequences A , B , and C , finds the length of their longest common subsequence. Briefly explain why your algorithm is correct and provide the runtime.
2. Rank the following functions in order of their asymptotic growth. That is if $f_i(n) = O(f_j(n))$ then $f_i(n)$ should come before $f_j(n)$ in your list. If $f_i(n) = \Theta(f_j(n))$ then the two functions should be given the same rank.
 - $f_1(n) = n^5$
 - $f_2(n) = n \log_2 n$
 - $f_3(n) = 10^{10}$
 - $f_4(n) = 2^{\log_2 n}$
 - $f_5(n) = 2n \log_{10} n$
 - $f_6(n) = (n + 1)!$
 - $f_7(n) = 2 \log_{10} n$
 - $f_8(n) = n^{\log_2 n}$
 - $f_9(n) = n(\log_2 n)^3$
3. Prove or disprove: Let $f(n)$ and $g(n)$ be functions that take on positive values for all $n > 1$ and such that $f(n) = O(g(n))$. Then $2^{f(n)} = O(2^{g(n)})$
4. Consider the following algorithm:

Algorithm 1 InsertionSort(A)

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1:  $n \leftarrow \text{length}(A)$ 
2: for  $i = 2$  to  $n$  do
3:    $x \leftarrow A[i]$ 
4:    $j \leftarrow i - 1$ 
5:   while  $j \geq 1$  and  $A[j] > x$  do
6:      $A[j + 1] \leftarrow A[j]$ 
7:      $j \leftarrow j - 1$ 
8:   end while
9:    $A[j + 1] \leftarrow x$ 
10: end for
```

Does this correctly sort an array A ? What's the best case running time? What's the worst case running time?